



Methods of Stochastic Modelling (Métodos de Modelação Estocástica)

Modelação e Desempenho de Redes e Serviços

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Random experiment

- A random experiment is an experiment whose outcome is uncertain
 - In a random experiment, the sample space, S , is the set of all possible results (i.e., outcomes) of the experiment
 - Any subset E of the sample space S is named an event
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- Given two events E and F :
 - The union of the events, $E \cup F$, is the set of possible results that belong to at least one of them
 - The intersection of the events, EF , is the set of possible results that belong to both events simultaneously
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- When $EF = \emptyset$ ($\emptyset$ is the empty set), we say that the events are mutually exclusive
 - The complement of an event E , E^c , is the set of all possible results (i.e., all results in the sample space S) that do not belong to E

Probabilities defined over events

- For each possible event E of a sample space S , the assignment of a real value $P(E)$ can represent the occurrence probability of event E , if it satisfies the following conditions:

$$(1) 0 \leq P(E) \leq 1$$

$$(2) P(S) = 1$$

$$(3) \text{ For any set of mutually exclusive events } E_1, E_2, E_3, \dots$$

$$P\left(\bigcup_i E_i\right) = \sum_i P(E_i)$$

- Corollaries:

$$P(E) + P(E^c) = 1$$

$$P(E \cup F) = P(E) + P(F) - P(EF)$$

Conditional probabilities

- Given two events E and F , the conditional probability of E knowing that F has occurred, represented by $P(E|F)$, is given by

$$P(E|F) = P(EF) / P(F)$$

- Two events E and F are said to be independent events if

$$P(EF) = P(E)P(F)$$

- If E and F are independent events, then:

$$P(E|F) = P(EF) / P(F) = P(E)P(F) / P(F) = P(E)$$

$$P(F|E) = P(FE) / P(E) = P(F)P(E) / P(E) = P(F)$$

meaning that the probability of each event is the same even if we know that the other event has occurred.

Law of Total Probability & Bayes Rule

Consider a set of n events $(F_1, F_2, \dots, F_n)$ that:

- are all mutually exclusive,
- their union is the sample space S of a random experiment.

Consider any other event E defined on the same random experiment.

Law of Total Probability: The probability of event E is given by:

$$P(E) = \sum_{i=1}^n P(EF_i) = \sum_{i=1}^n P(E | F_i)P(F_i)$$

Bayes Rule: Knowing that event E has occurred, the occurrence probability of event F_j (with $j = 1, 2, \dots, n$) is given by:

$$P(F_j | E) = \frac{P(EF_j)}{P(E)} = \frac{P(E | F_j)P(F_j)}{P(E)} = \frac{P(E | F_j)P(F_j)}{\sum_{i=1}^n P(E | F_i)P(F_i)}$$

Example 1

In a wireless link between two hosts, the probability of the transmitted data packets being received with errors is 0.1% in normal link conditions or 10% with external interference. The probability of external interference is 2%. In reception, the hosts are able to detect with 100% probability if each data packet is or is not received with errors.

Determine: (i) the probability of a data packet being received with errors and (ii) the probability of the link being with interference when a data packet is received with errors.

Events:

- E – the packet is received with errors
- F_1 – the link is in the normal state
- F_2 – the link is with interference

(i) Law of Total Probability

$$\begin{aligned} P(E) &= P(E|F_1)P(F_1) + P(E|F_2)P(F_2) \\ &= 0.001 \times (1 - 0.02) + 0.1 \times 0.02 \\ &= 0.00298 = 0.298\% \end{aligned}$$

(ii) Bayes Rule

$$\begin{aligned} P(F_2|E) &= P(E|F_2)P(F_2) / P(E) \\ &= 0.1 \times 0.02 / 0.00298 \\ &= 0.671 = 67.1\% \end{aligned}$$

Random variables

- A random variable X is a function that assigns a real value to each possible result in S of a random experiment.
- The distribution function (or *cumulative distribution function*) of a random variable X is defined as:

$$F(x) = P(X \leq x) \quad , \quad -\infty < x < +\infty$$

- Properties of the distribution function:

(1) $0 \leq F(x) \leq 1$ for all values of x

(2) if $x_1 \leq x_2$ then $F(x_1) \leq F(x_2)$ (non-decreasing function)

(3) $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow +\infty} F(x) = 1$

(4) $P(a < X \leq b) = F(b) - F(a)$, for $a < b$

Discrete random variables

- X is a discrete random variable if it can only take a finite number of values or a countable number of distinct separate values $x_1, x_2, \dots, x_i, \dots$
- The probability function (or *mass probability function*) of a discrete random variable X is defined as:

$$f(x_i) = P(X = x_i) \quad \text{for all values of } i = 1, 2, 3, \dots$$

It is mandatory that: $\sum_{i=1}^{\infty} f(x_i) = 1$

- Consequently, the distribution function of X becomes:

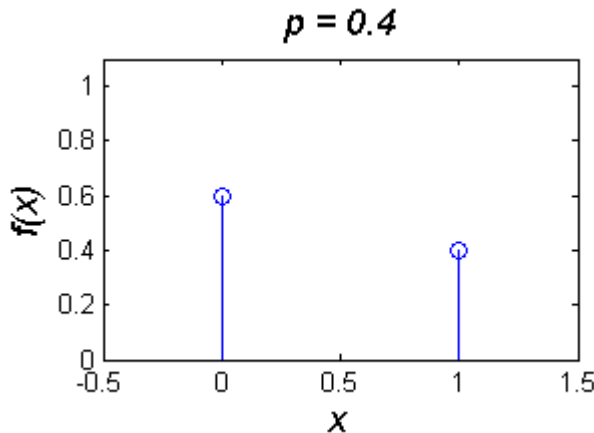
$$F(x) = \sum_{x_i \leq x} f(x_i) \quad , \quad -\infty < x < +\infty$$

Examples of discrete random variables

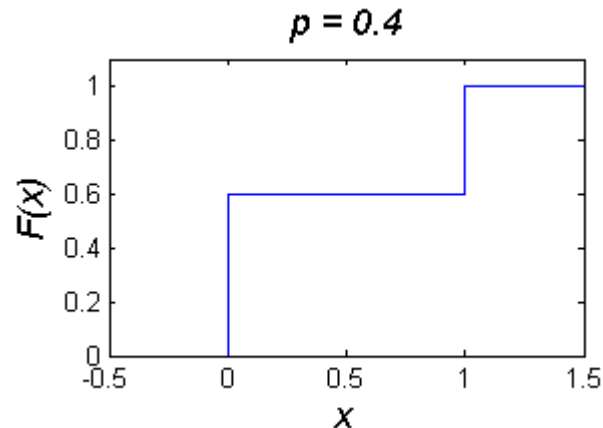
Bernoulli random variable represents an experiment that can be a success with probability p or a failure with probability $1 - p$.

If $X = 1$ represents the success and $X = 0$ the failure, the probability function is:

$$f(i) = p^i (1 - p)^{1-i}, i = 0, 1$$



$f(x)$ – probability function



$F(x)$ – distribution function

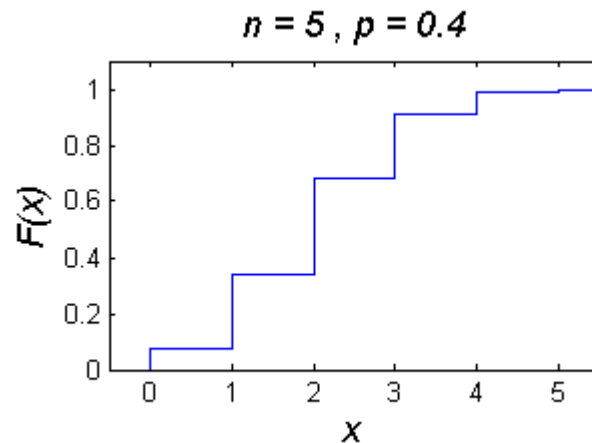
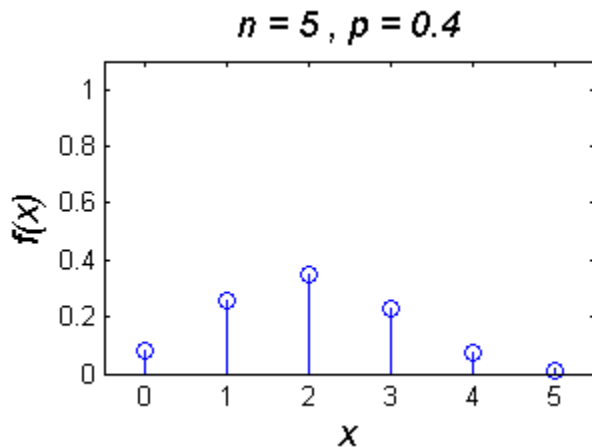
Examples of discrete random variables

Binomial random variable represents a set of n independent Bernoulli experiments, each one that can be a success with probability p or a failure with probability $1 - p$.

If X represent the number of successes in n experiments, the probability function is:

$$f(i) = \binom{n}{i} p^i (1-p)^{n-i}, i = 0, 1, 2, \dots, n$$

where $\binom{n}{i} = \frac{n!}{i!(n-i)!}$

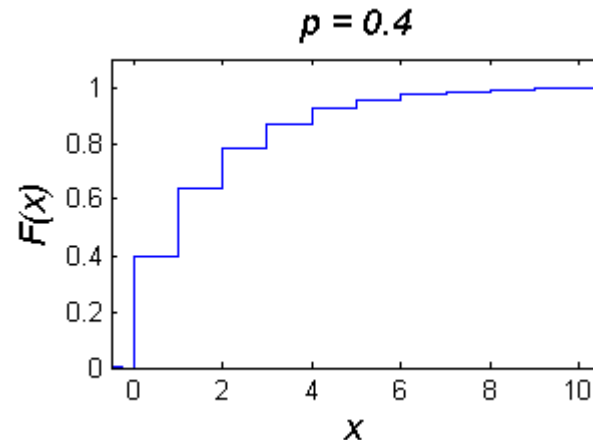
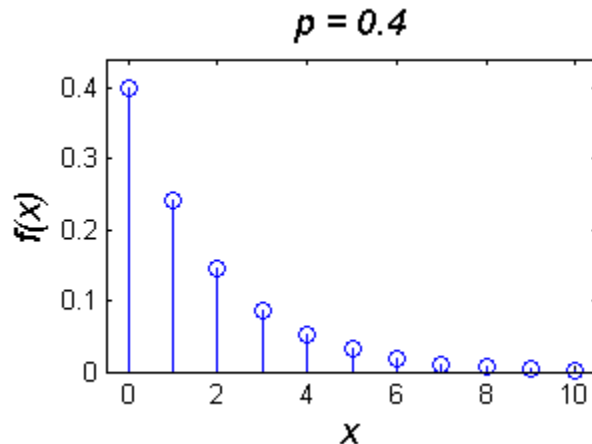


Examples of discrete random variables

Geometric random variable represents a set independent Bernoulli experiments, all with the same success probability, until an experiment results in a success.

If X represents the number of failures before the success, the probability function is

$$f(i) = (1-p)^i p, \quad i = 0, 1, 2, \dots$$



If X represents the number of experiments until the success, the probability function is

$$f(i) = (1-p)^{i-1} p, \quad i = 1, 2, \dots$$

Example 2

On a given data link, the BER (*bit error rate*) is 10^{-5} and the errors in the different bits of a data packet are statistically independent.

Determine: (i) the probability of a data packet of size 100 Bytes to be received without errors and (ii) the probability of a data packet of size 1000 Bytes to be received with at least 2 errors.

The number of bits in error on a data packet is a binomial random variable with the probability of success given by the BER value and the number of Bernoulli experiments given by the number of bits of the packet

$$f(i) = \binom{n}{i} p^i (1-p)^{n-i}, i = 0, 1, 2, \dots, n$$

$$(i) \quad f(0) = \binom{n}{0} p^0 (1-p)^{n-0} = \binom{100 \times 8}{0} \times (1-10^{-5})^{100 \times 8} = 0.992 = 99.2\%$$

$$(ii) \quad 1 - f(0) - f(1) = 1 - \binom{n}{0} p^0 (1-p)^{n-0} - \binom{n}{1} p^1 (1-p)^{n-1} \\ = 1 - (1-10^{-5})^{8000} - 8000 \times 10^{-5} (1-10^{-5})^{7999} = 3.034\text{E} - 3 = 0.3\%$$

Continuous random variables

- A random variable X is said continuous if it exists a non-negative function $f(x)$ such that for any interval of continuous values B :

$$P(X \in B) = \int_B f(x) dx$$

It is mandatory that:
$$\int_{-\infty}^{+\infty} f(x) dx = 1$$

$f(x)$ is the probability density function of the random variable X

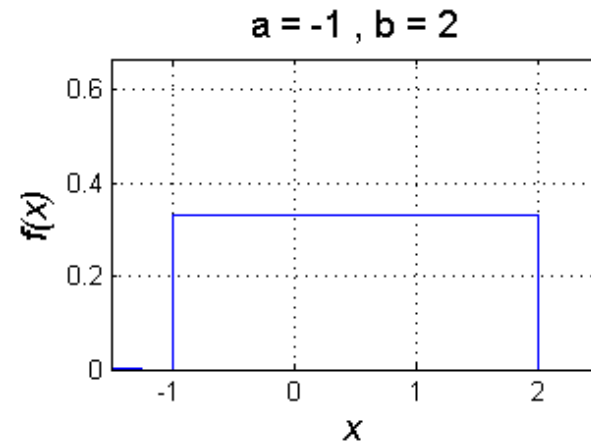
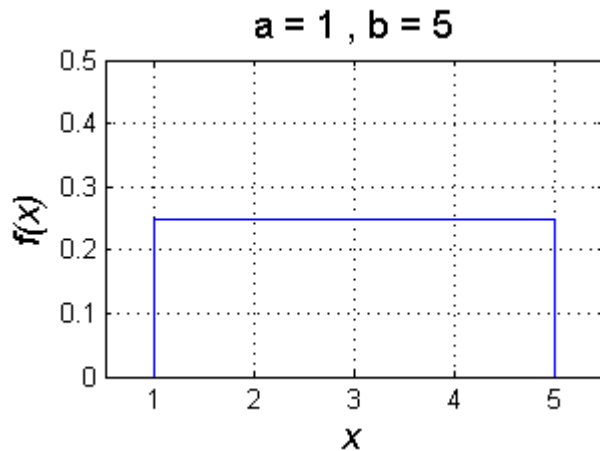
- Therefore:
$$P\{a \leq X \leq b\} = \int_a^b f(x) dx$$
- The distribution function of the random variable X becomes:

$$F(x) = P(X \in [-\infty, x]) = \int_{-\infty}^x f(y) dy$$

Examples of continuous random variables

Random variable with uniform distribution: a random variable is uniformly distributed in the interval $[a,b]$ if its probability density function is given by

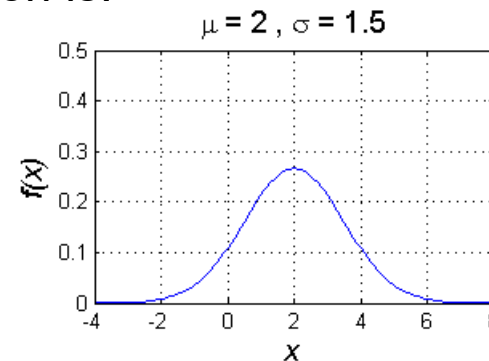
$$f(x) = \begin{cases} \frac{1}{b-a} & , a < x < b \\ 0 & , \text{cc} \end{cases}$$



Examples of continuous random variables

Random variable with Normal (ou Gaussian) distribution: a random variable has a normal distribution with average μ and standard deviation σ if its probability density function is:

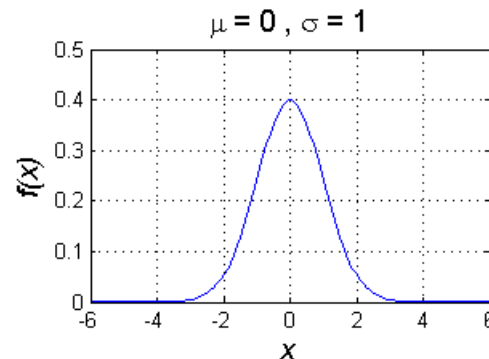
$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}$$



The Standard Normal (or Gaussian) distribution is the normal distribution with average 0 and standard deviation 1.

In this case:

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$



Average (or expected value) of a random variable

- Average (or expected value), $E[X]$, of a random variable X :

$$E[X] = \begin{cases} \sum_{j=1}^{\infty} x_j f_X(x_j) & \text{if } X \text{ discrete} \\ \int_{-\infty}^{+\infty} x f_X(x) dx & \text{if } X \text{ continuous} \end{cases}$$

- Important property of the average: $E\left[\sum_{i=1}^n c_i X_i\right] = \sum_{i=1}^n c_i E[X_i]$

- Average of a random variable $Y = g(X)$:

$$E[g(X)] = \begin{cases} \sum_{j=1}^{\infty} g(x_j) f_X(x_j) & \text{if } X \text{ discrete} \\ \int_{-\infty}^{+\infty} g(x) f_X(x) dx & \text{if } X \text{ continuous} \end{cases}$$

Variance and standard deviation of a random variable

- Variance of a random variable X :

$$\text{Var}[X] = E\left[\left(X - E[X]\right)^2\right] = E[X^2] - E[X]^2$$

- Important properties of the variance:

2nd moment of X

$$\text{Var}[X] \geq 0$$

$$\text{Var}[cX] = c^2 \text{Var}[X]$$

$$\text{Var}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \text{Var}[X_i] \quad \text{if } X_i \text{ are independent}$$

- Standard deviation of a random variable X :

$$\sigma[X] = \sqrt{\text{Var}[X]}$$

Example 3

A data link of 10 Mbps supports a flow of data packets whose size is 100 Bytes with probability 10%, 500 Bytes with probability 50% and 1500 Bytes with probability 40%. Consider the random variable X representing the packet transmission time.

Determine: (i) the average packet transmission time $E[X]$, (ii) the second moment of the packet transmission time $E[X^2]$ and (iii) the variance of the packet transmission time $Var[X]$.

$$(i) \quad E[X] = \sum_{j=1}^{\infty} x_j f_X(x_j) = \frac{100 \times 8}{10^7} \times 0.1 + \frac{500 \times 8}{10^7} \times 0.5 + \frac{1500 \times 8}{10^7} \times 0.4$$
$$= 0.688 \times 10^{-3} \text{ sec} = 0.688 \text{ msec}$$

$$(ii) \quad E[X^2] = \sum_{j=1}^{\infty} (x_j)^2 f_X(x_j) = \left(\frac{100 \times 8}{10^7} \right)^2 \times 0.1 + \left(\frac{500 \times 8}{10^7} \right)^2 \times 0.5 + \left(\frac{1500 \times 8}{10^7} \right)^2 \times 0.4$$
$$= 6.5664 \times 10^{-7} \text{ sec}^2$$

Example 3 - continuation

A data link of 10 Mbps supports a flow of data packets whose size is 100 Bytes with probability 10%, 500 Bytes with probability 50% and 1500 Bytes with probability 40%. Consider the random variable X representing the packet transmission time.

Determine: (i) the average packet transmission time $E[X]$, (ii) the second moment of the packet transmission time $E[X^2]$ and (iii) the variance of the packet transmission time $Var[X]$.

(iii) **1st alternative:** $Var[X] = E\left[\left(X - E[X]\right)^2\right]$

$$\begin{aligned} Var[X] &= \left(\frac{100 \times 8}{10^7} - E[X]\right)^2 \times 0.1 + \left(\frac{500 \times 8}{10^7} - E[X]\right)^2 \times 0.5 + \left(\frac{1500 \times 8}{10^7} - E[X]\right)^2 \times 0.4 \\ &= 1.833 \times 10^{-7} \text{ sec}^2 \end{aligned}$$

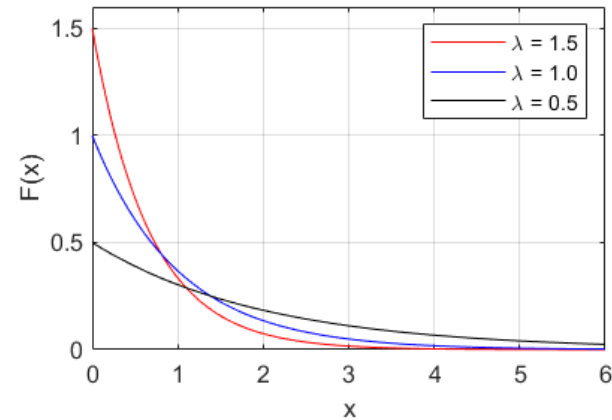
2nd alternative: $Var[X] = E[X^2] - E[X]^2$

$$Var[X] = 6.5664 \times 10^{-7} - (0.688 \times 10^{-3})^2 = 1.833 \times 10^{-7} \text{ sec}^2$$

Random variable with exponential distribution

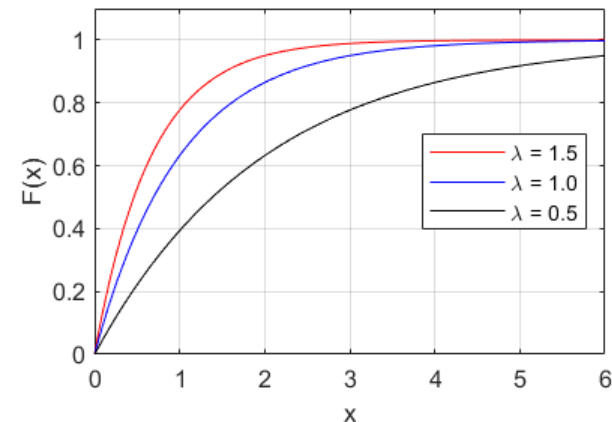
- A continuous random variable X following an exponential distribution with parameter λ , $\lambda > 0$, has the probability density function:

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$



- The distribution function is:

$$F(x) = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$



Exponential distribution

The exponential distribution has the following properties:

1. The average, variance and standard deviation of an exponentially distributed random variable X are only dependent on parameter λ :

$$E[X] = \frac{1}{\lambda} \quad Var[X] = E[X]^2 = \left(\frac{1}{\lambda}\right)^2 \quad \sigma[X] = E[X] = \frac{1}{\lambda}$$

2. The exponential distribution has no memory, meaning that:

$$P\{X > s + t \mid X > t\} = P\{X > s\}$$

3. If n random variables $(X_1, X_2, \dots, X_n)$ are all independent and exponentially distributed (with averages $1/\lambda_1, 1/\lambda_2, \dots, 1/\lambda_n$ respectively), the probability of the value of X_i (with $i = 1, 2, \dots, n$) being the lowest among the values of all variables is:

$$\frac{\lambda_i}{\lambda_1 + \lambda_2 + \dots + \lambda_n}$$

Example 4

A data link of 10 Mbps supports a flow of data packets whose size is exponentially distributed with an average of 1000 Bytes. Consider the random variable X representing the packet transmission time.

Determine: (i) the average packet transmission time $E[X]$, (ii) the variance of the packet transmission time $Var[X]$ and (iii) the second moment of the packet transmission time $E[X^2]$.

(i) $E[X] = \frac{1000 \times 8}{10^7} = 8 \times 10^{-4} = 0.8 \text{ msec}$ Capacity of the link in pps (packets per second)

$$E[X] = \frac{1}{\mu} \Leftrightarrow \mu = \frac{1}{E[X]} = \frac{1}{8 \times 10^{-4}} = 1250 \text{ pps}$$

(ii) $Var[X] = \left(\frac{1}{\mu}\right)^2 = (8 \times 10^{-4})^2 = 6.4 \times 10^{-7} \text{ sec}^2$

(iii) $Var[X] = E[X^2] - E[X]^2 \Leftrightarrow E[X^2] = Var[X] + E[X]^2$

$$E[X^2] = 6.4 \times 10^{-7} + (8 \times 10^{-4})^2 = 1.28 \times 10^{-6} \text{ sec}^2$$

Stochastic process

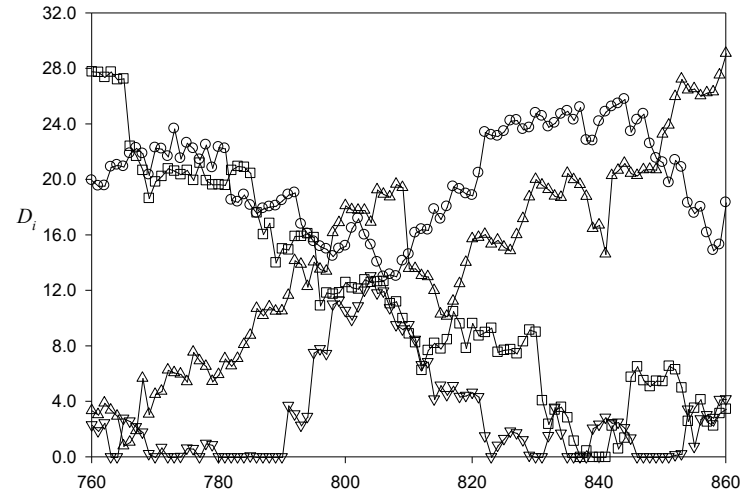
- A stochastic process $\{X(t), t \in T\}$ is a set of random variables: for each value $t \in T$, $X(t)$ is a random variable.
- Index t is frequently interpreted as a time instant. In this interpretation, the random variable $X(t)$ represents the state of the stochastic process on time instant t .
- Set T is the set of all possible indices of the stochastic process:
 - (1) if T is a countable set, the stochastic process is designated as being in discrete time
 - (2) if T is an interval of continuous values, the stochastic process is designated as being in continuous time
- The state space of the stochastic process is the set of all possible values that the random variables $X(t)$ can take.

Examples of stochastic processes

Consider a system with a queue and a server. Clients arrive to the system and are either immediately served (if the server is empty) or go to the queue to wait to be served.

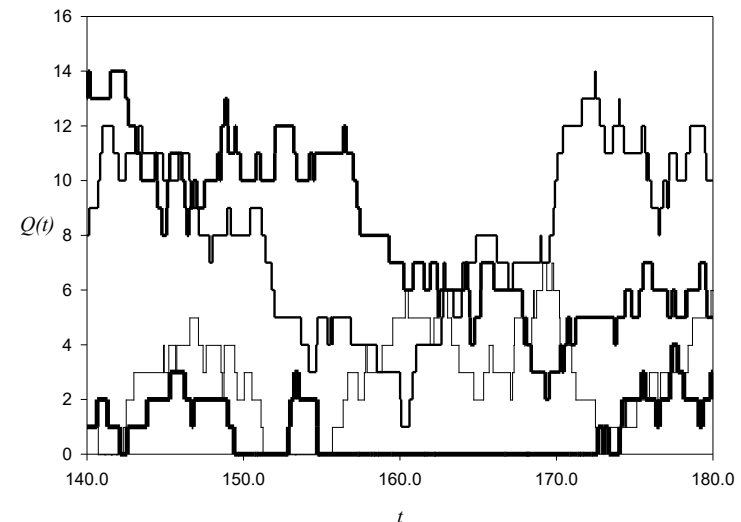
Queuing delay of clients

- (1) is a stochastic process in discrete time (1st client, 2nd client, ...)
- (2) the state is a continuous value (queuing delay value continuous)



Number of clients in the queue

- (1) is a stochastic process in continuous time
- (2) the state is a discrete value (0 clients, 1 client, 2 clients, ...)



Counting process

- A stochastic process $\{N(t), t \geq 0\}$ is a counting process if $N(t)$ represents the total number of events occurred until time instant t .
- A counting process satisfies the following conditions:
 - (1) $N(t) \geq 0$.
 - (2) $N(t)$ takes only non-negative integer values.
 - (3) If $s < t$, then $N(s) \leq N(t)$.
 - (4) If $s < t$, then $N(t) - N(s)$ is equal to the number of events occurred in the time interval $[s, t]$.
- A counting process has independent increments if the number of events in disjoint time intervals is independent.
- A counting process has stationary increments if the distribution function of the number of events occurred in any time interval is only dependent on the time interval length.

Poisson process

- A counting process is a Poisson process with rate λ , $\lambda > 0$, if the following conditions hold:
 - (1) $N(0) = 0$;
 - (2) the process has independent increments;
 - (3) the number of events in a time interval of length t has a Poisson distribution with average λt , i.e., for all $s, t \geq 0$

$$P\{N(s+t) - N(s) = n\} = e^{-\lambda t} \frac{(\lambda t)^n}{n!}$$

- A Poisson process has stationary increments and the average number of events in a time interval of length t is:

$$E[N(t)] = \lambda t$$

the reason why the parameter λ is named the rate (i.e., the average number of events per time unit) of the Poisson process.

Properties of a Poisson process

- **Property 1:** Consider in a Poisson process with rate λ that:
 - T_1 is the time instant of the first event
 - $T_n, n > 1$, is the time interval length between the $(n-1)^{\text{th}}$ event and the n^{th} event
- Then, $T_n, n = 1, 2, \dots$, are independent and identically distributed random variables with an exponential distribution with average $1/\lambda$.
- **Property 2:** Consider in a Poisson process $\{N(t), t \geq 0\}$ with rate λ that each event is independently classified as:
 - event of type 1 with probability p
 - event of type 2 with probability $1 - p$

and that $\{N_1(t), t \geq 0\}$ and $\{N_2(t), t \geq 0\}$ represent the number of events of each type occurred in the time interval $[0, t]$.

- Then, $N_1(t)$ e $N_2(t)$ are both independent Poisson processes with rates λp and $\lambda(1-p)$, respectively.

Properties of a Poisson process

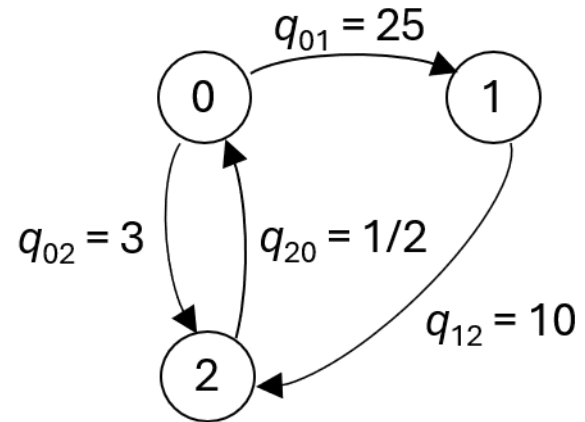
- **Property 3:** Consider $\{N_1(t), t \geq 0\}$ and $\{N_2(t), t \geq 0\}$ as two independent Poisson processes with rates λ_1 and λ_2 , respectively.
- Then, the process $N(t) = N_1(t) + N_2(t)$ is also a Poisson process with rate $\lambda = \lambda_1 + \lambda_2$.
- **Property 4:** Consider that we know that exactly n events have occurred from the beginning until time instant t in a Poisson process.
- Then, the occurrence time instants of the n events are independent random variables with a uniform distribution in the interval $[0, t]$.

Continuous-Time Markov Chain (CTMC)

- A Continuous-Time Markov Chain (CTMC) is a type of stochastic process that models a system moving between states randomly at any continuous moment in time.
- A CTMC is characterized by the following properties:
 - **Markovian property (Memorylessness):** The future behaviour of the system depends only on its current state. The past of the system is irrelevant to predicting what happens next.
 - **Holding time in each state is exponentially distributed:** Because the process is memoryless, the holding time at each state (i.e., the amount of time the system stays in each state before moving to another state) follows an exponential distribution (the only continuous probability distribution entirely memoryless).
 - **Transition Rates:** A CTMC is defined by a set of states and a set of transition rates between states, which define how fast the system is likely to move from each state to each of the other states.

Continuous-Time Markov Chain (CTMC)

- Consider CTMC with a state space defined by non-negative integer values (i.e., 0, 1, 2, ...).
- Consider that the rate from state i to state j is given by q_{ij} .
- Example (rates in transitions per second):

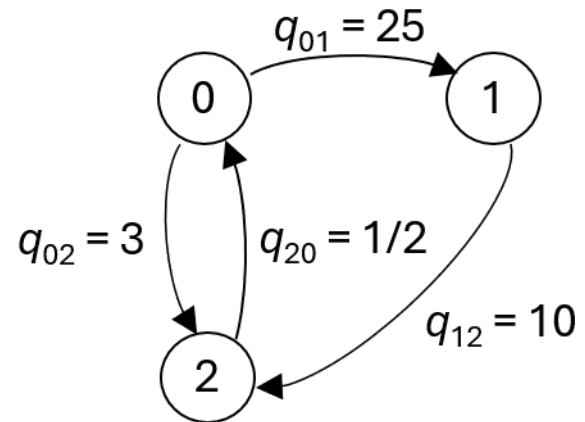


This CTMC has 3 states:

- when the system enters state 0, it moves either to state 1 (at a rate of 25 transitions per second) or to state 2 (at a rate of 3 transitions per second),
- when the system enters state 1, it moves always to state 2 (at a rate of 10 transitions per second),
- when the system enters state 2, it moves always to state 0 (at a rate of 1/2 transitions per second).

Continuous-Time Markov Chain (CTMC)

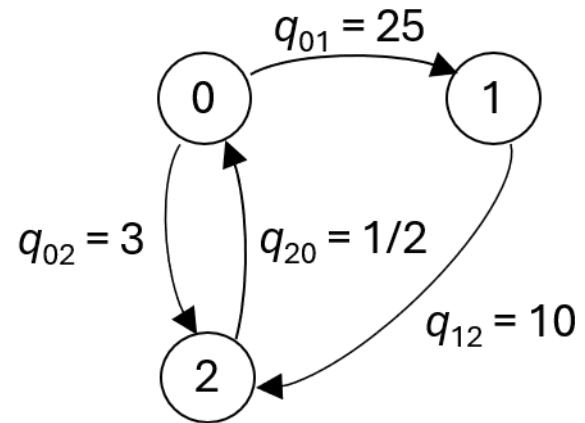
- Example (rates in transitions per second):



- The average holding time on each state is the inverse of the rate at which the system moves from it to the other state:
 - **State 0:** Average holding time is $1/(25+3) = 0.0357$ seconds
 - **State 1:** Average holding time is $1/10 = 0.1$ seconds
 - **State 2:** Average holding time is $1/(1/2) = 2$ seconds

Continuous-Time Markov Chain (CTMC)

- Example (rates in transitions per second):



- At each state, the probability of the next state is proportional to the transition rate to each possible next state:
 - **State 0:** Next state is state 1 with probability $25/(25+3) = 89.3\%$ or state 2 with $3/(25+3) = 10.7\%$
 - **State 1:** Next state is state 2 with probability 100%
 - **State 2:** Next state is state 0 with probability 100%

State limit probabilities of a CTMC

- Consider $P_{ij}(t) = P\{X(s+t) = j \mid X(s) = i\}$

the probability of state j after a time duration t when the CTMC is in state i at the present time.

- The probability of a CTMC being on state j converges to a limit value π_j which is independent of the initial state:

$$\pi_j \equiv \lim_{t \rightarrow \infty} P_{ij}(t)$$

- For a CTMC to have state limit probabilities π_j , it must be:
 - irreducible (all states can reach each other),
 - aperiodic (there's no fixed cycle of states),
 - positive recurrent (starting on any state, the average time to return to it is finite to all states).
- In this course unit, all CTMCs of interest have these properties.

Computing state limit probabilities of CTMC

- The state limit probabilities π_j can be computed by resolving the following set of equations:

$$q_j \pi_j = \sum_{k \neq j} q_{kj} \pi_k, \quad \text{for all states } j$$

$$\sum_j \pi_j = 1$$

- These equations are known as the balance equations:

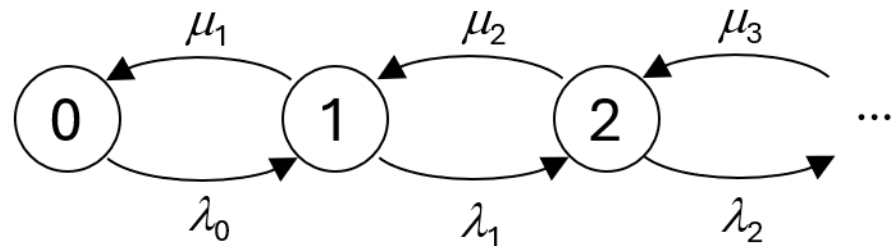
rate at which the system transits from state j to another state
=
rate at which the system transits from any other state to state j

- The probability π_j gives also the percentage of time that the process is in state j .
- The state limit probabilities are also named steady-state probabilities.

Birth-dead process

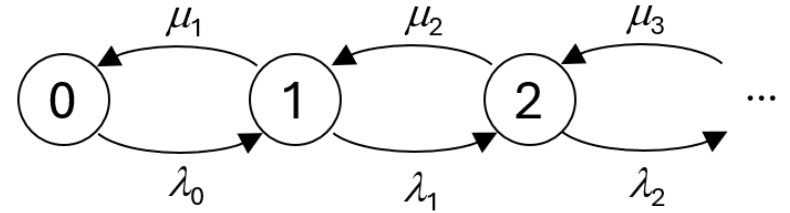
- Consider a CTMC whose state represents the number of clients, and the system has an infinite capacity (i.e., the system can accommodate an infinite number of clients).
- When the system is in state n (i.e., it has n clients):
 - (1) new clients arrive to the system at an exponential rate λ_n
 - (2) clients leave the system at an exponential rate μ_n

State transition diagram:



- This CTMC is named a birth-death process.
- Parameters λ_n ($n = 0, 1, \dots$) and μ_n ($n = 1, 2, \dots$) are referred to as the birth rates and the death rates, respectively.

State limit probabilities of birth-death processes



Limit probability π_n of state n (solving the balance equations):

$$\pi_0 = \frac{1}{1 + \frac{\lambda_0}{\mu_1} + \left(\frac{\lambda_0}{\mu_1} \times \frac{\lambda_1}{\mu_2}\right) + \left(\frac{\lambda_0}{\mu_1} \times \frac{\lambda_1}{\mu_2} \times \frac{\lambda_2}{\mu_3}\right) + \dots} = \frac{1}{1 + \sum_{i=1}^{\infty} \left(\frac{\lambda_0 \lambda_1 \dots \lambda_{i-1}}{\mu_1 \mu_2 \dots \mu_i}\right)}$$

$$\pi_n = \frac{\frac{\lambda_0}{\mu_1} \times \frac{\lambda_1}{\mu_2} \times \dots \times \frac{\lambda_{n-1}}{\mu_n}}{1 + \frac{\lambda_0}{\mu_1} + \left(\frac{\lambda_0}{\mu_1} \times \frac{\lambda_1}{\mu_2}\right) + \left(\frac{\lambda_0}{\mu_1} \times \frac{\lambda_1}{\mu_2} \times \frac{\lambda_2}{\mu_3}\right) + \dots} = \frac{\lambda_0 \lambda_1 \dots \lambda_{n-1}}{\mu_1 \mu_2 \dots \mu_n} \pi_0, \quad n = 1, 2, 3, \dots$$

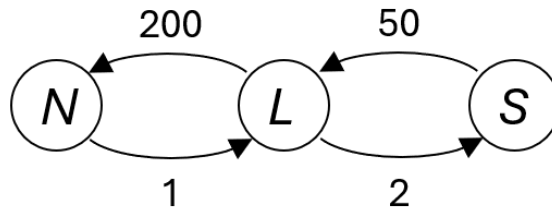
NOTE: If the process has a finite number N of states (i.e., $n = 0, 1, \dots, N$), the summations in the above expressions are from 0 to N .

In the case of an infinite number of states, the state limit probabilities exist only if:

$$\sum_{i=1}^{\infty} \left(\frac{\lambda_0 \lambda_1 \dots \lambda_{i-1}}{\mu_1 \mu_2 \dots \mu_i}\right) < \infty$$

Example 5

Consider a wireless link for packet transmission that can be in one of 3 possible states – Normal (N), Light Interference (L) or Severe Interferences (S) – according with the following Markov chain (rates in transitions per hour):

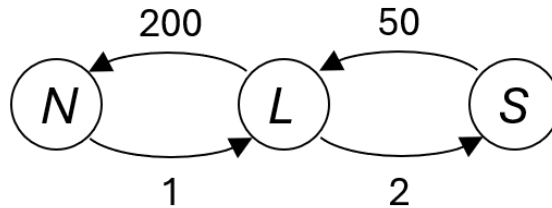


The probability of a data packet being received with one or more errors is 0.01% in state N , 0.1% in state L and 1% in state S . Determine:

- (a) the probability of each state,
- (b) the average holding time (in minutes) of the link in each state,
- (c) the probability of the data packet being received with errors,
- (d) the probability of state N when a data packet is received with errors.

Example 5 – solution of (a)

Consider a wireless link for packet transmission that can be in one of 3 possible states – Normal (N), Light Interference (L) or Severe Interferences (S) – according with the following Markov chain (rates in transitions per hour):



The probability of a data packet being received with one or more errors is 0.01% in state N , 0.1% in state L and 1% in state S . Determine:

(a) the probability of each state,

$$P_N = \frac{1}{1 + \frac{1}{200} + \frac{1}{200} \times \frac{2}{50}} = 0.99483 = 99.483\%$$

$$P_L = \frac{\frac{1}{200}}{1 + \frac{1}{200} + \frac{1}{200} \times \frac{2}{50}} = 0.00497 = 0.497\%$$

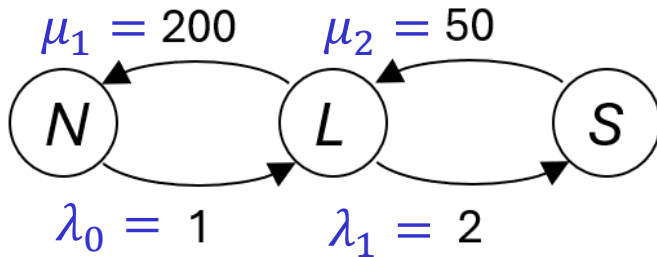
$$P_S = \frac{\frac{1}{200} \times \frac{2}{50}}{1 + \frac{1}{200} + \frac{1}{200} \times \frac{2}{50}} = 0.0002 = 0.02\%$$

$$\pi_0 = \frac{1}{1 + \sum_{i=1}^{\infty} \left(\frac{\lambda_0 \lambda_1 \dots \lambda_{i-1}}{\mu_1 \mu_2 \dots \mu_i} \right)}$$

$$\pi_n = \frac{\lambda_0 \lambda_1 \dots \lambda_{n-1}}{\mu_1 \mu_2 \dots \mu_n} \pi_0$$

Example 5 – solution of (a) in MATLAB

How to calculate the state limit probabilities in MATLAB:



$$\pi_0 = \frac{1}{1 + \sum_{i=1}^{\infty} \left(\frac{\lambda_0 \lambda_1 \dots \lambda_{i-1}}{\mu_1 \mu_2 \dots \mu_i} \right)}$$

$$\pi_n = \frac{\lambda_0 \lambda_1 \dots \lambda_{n-1}}{\mu_1 \mu_2 \dots \mu_n} \pi_0$$

```
lambda= [1 2];
```

```
miu= [200 50];
```

```
co= [1 , lambda./miu];
```

```
co= cumprod(co);
```

```
u= co/sum(co);
```

```
fprintf('Prob. of state N: %.2e\n',u(1));
```

```
fprintf('Prob. of state L: %.2e\n',u(2));
```

```
fprintf('Prob. of state S: %.2e\n',u(3));
```

$$\left[1, \frac{\lambda_0}{\mu_1}, \frac{\lambda_1}{\mu_2} \right]$$

$$\left[1, 1 \times \frac{\lambda_0}{\mu_1}, 1 \times \frac{\lambda_0}{\mu_1} \times \frac{\lambda_1}{\mu_2} \right]$$

$$\left[\frac{1}{1 + \frac{\lambda_0}{\mu_1} + \frac{\lambda_0 \lambda_1}{\mu_1 \mu_2}}, \frac{\frac{\lambda_0}{\mu_1}}{1 + \frac{\lambda_0}{\mu_1} + \frac{\lambda_0 \lambda_1}{\mu_1 \mu_2}}, \frac{\frac{\lambda_0 \lambda_1}{\mu_1 \mu_2}}{1 + \frac{\lambda_0}{\mu_1} + \frac{\lambda_0 \lambda_1}{\mu_1 \mu_2}} \right]$$

Result:

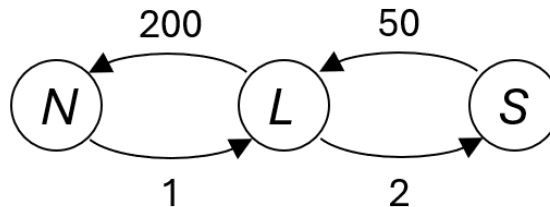
Prob. of state N: 9.95e-01

Prob. of state L : 4.97e-03

Prob. of state S : 1.99e-04

Example 5 – solution of (b)

Consider a wireless link for packet transmission that can be in one of 3 possible states – Normal (N), Light Interference (L) or Severe Interferences (S) – according with the following Markov chain (rates in transitions per hour):



The probability of a data packet being received with one or more errors is 0.01% in state N , 0.1% in state L and 1% in state S . Determine:

(b) the average holding time (in minutes) of the link in each state,

$$T_N = \frac{1}{1} = 1 \text{ hour} = 60 \text{ minutes}$$

$$T_L = \frac{1}{2 + 200} = 0.00495 \text{ hours} = 0.3 \text{ minutes}$$

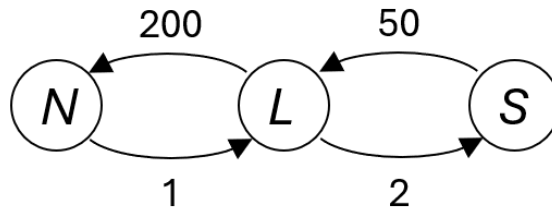
$$T_S = \frac{1}{50} = 0.02 \text{ hours} = 1.2 \text{ minutes}$$

Average holding time
of state i :

$$T_i = \frac{1}{\sum_{j \neq i} q_{ij}}$$

Example 5 – solution of (c)

Consider a wireless link for packet transmission that can be in one of 3 possible states – Normal (N), Light Interference (L) or Severe Interferences (S) – according with the following Markov chain (rates in transitions per hour):



The probability of a data packet being received with one or more errors is 0.01% in state N , 0.1% in state L and 1% in state S . Determine:

(c) the probability of the data packet being received with errors,

Law of total probability

$$P(E) = P(E|N) P(N) + P(E|L) P(L) + P(E|S) P(S)$$

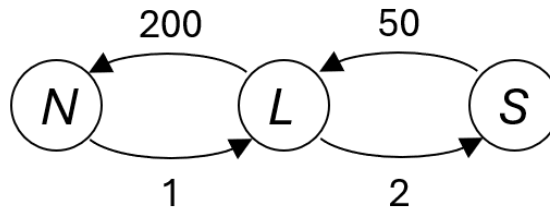
Calculated in 5(a)

$$= 0.0001 \times 0.99483 + 0.001 \times 0.00497 + 0.01 \times 0.0002$$

$$= 1.0445 \times 10^{-4} = 0.010445\%$$

Example 5 – solution of (d)

Consider a wireless link for packet transmission that can be in one of 3 possible states – Normal (N), Light Interference (L) or Severe Interferences (S) – according with the following Markov chain (rates in transitions per hour):



The probability of a data packet being received with one or more errors is 0.01% in state N , 0.1% in state L and 1% in state S . Determine:

(d) the probability of state N when a data packet is received with errors.

Bayes Rule

$$\begin{aligned}
 P(N|E) &= \frac{P(E|N) P(N)}{P(E|N) P(N) + P(E|L) P(L) + P(E|S) P(S)} \\
 &= \frac{0.0001 \times 0.99483}{0.0001 \times 0.99483 + 0.001 \times 0.00497 + 0.01 \times 0.0002} \\
 &= 0.9346 = 93.46\%
 \end{aligned}$$

Calculated
in 5(a)